

northwestern Mexico (**Figure 52.17**). To the north, their range is limited by an abiotic factor: temperature. Saguaros tolerate freezing temperatures only briefly, typically for less than a day, and generally cannot survive at temperatures below -4°C (25°F). For the same reason, saguaros are rarely found at elevations above 1,200 m (4,000 feet).

However, temperature alone does not fully explain the distribution of saguaros, including why they are missing from the western portion of the Sonoran Desert. Water availability is critical because seedling survival typically requires consecutive years of moist conditions, something that may occur only a few times each century. Biotic factors almost certainly influence their distribution as well. Mice and grazers such as goats eat the seedlings, and bats pollinate the large, white flowers that open at night. Saguaros are also vulnerable to a deadly bacterial disease. Thus, for the saguaro, as for most other species, ecologists need to consider multiple factors and alternative hypotheses when attempting to explain the distribution of a species.

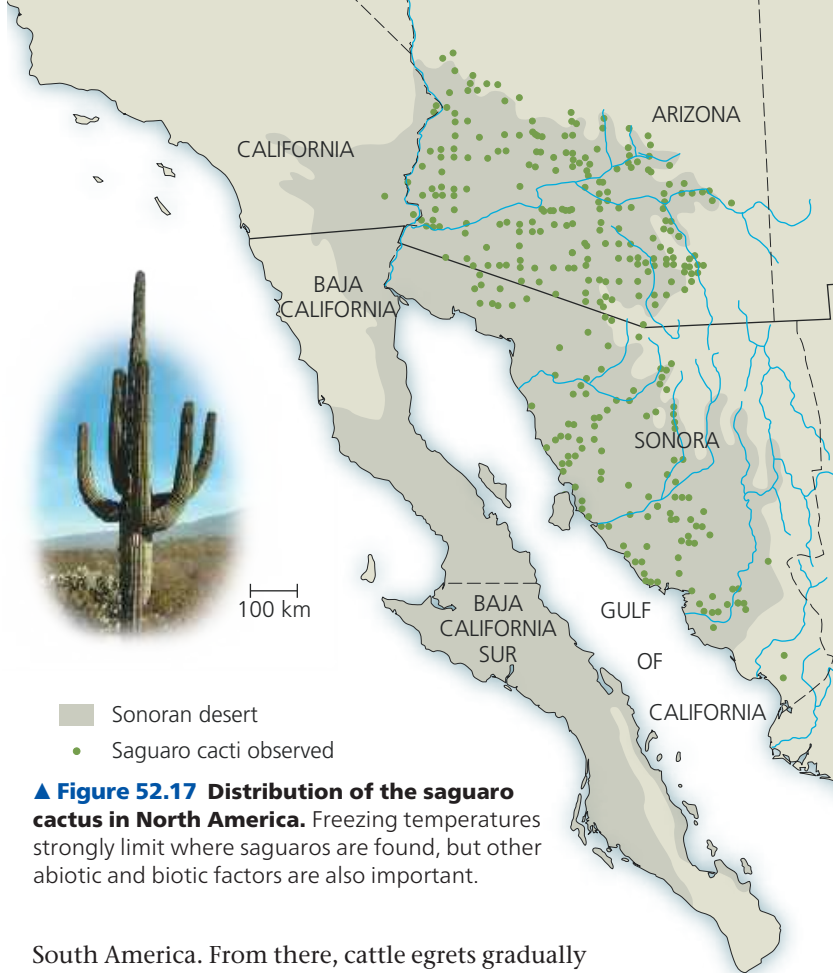
To see how ecologists might arrive at such an explanation, let's examine the ecological factors highlighted by the questions in the flowchart in **Figure 52.18**.

Dispersal and Distribution

One factor that contributes greatly to the global distribution of organisms is **dispersal**, the movement of individuals or gametes away from their area of origin or from centers of high population density. For example, while land-bound kangaroos have not reached Africa under their own power, other organisms that disperse more readily, such as some birds, have. The dispersal of organisms is critical to understanding the role of geographic isolation in evolution (see Concept 24.2) as well as the patterns of species distribution we see today, including that of the Pacific diatom discussed earlier in this chapter.

Natural Range Expansions and Adaptive Radiation

The importance of dispersal is most evident when organisms reach an area where they did not exist previously, called a *range expansion*. For instance, 200 years ago, the cattle egret (*Bubulcus ibis*) was found only in Africa and southwestern Europe. But in the late 1800s, some of these birds managed to cross the Atlantic Ocean and colonize northeastern



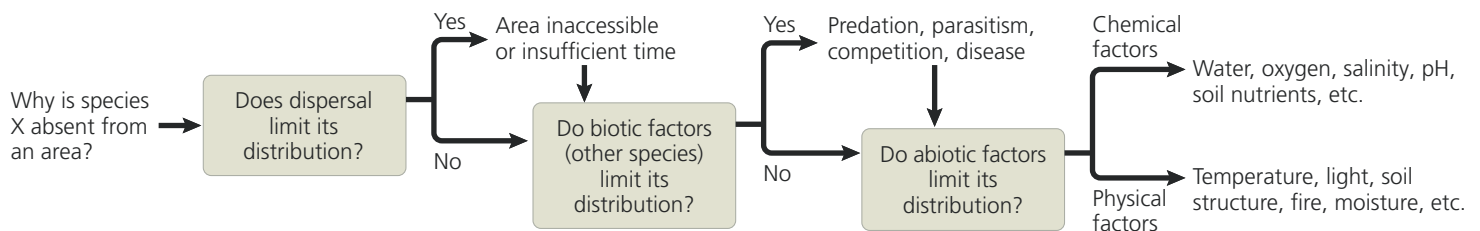
▲ Figure 52.17 Distribution of the saguaro cactus in North America. Freezing temperatures strongly limit where saguaros are found, but other abiotic and biotic factors are also important.

South America. From there, cattle egrets gradually spread southward and also northward through Central America and into North America, reaching Florida by 1960 (**Figure 52.19**). Today they have breeding populations as far west as the Pacific coast of the United States and as far north as southern Canada.

In rare cases, such long-distance dispersal can lead to adaptive radiation, the rapid evolution of an ancestral species into new species that fill many ecological niches. The diversity of Hawaiian silverswords is one example of adaptive radiation that was possible only with the long-distance dispersal of an ancestral tarweed from North America (see Figure 25.22).

Natural range expansions clearly show the influence of dispersal on distribution. However, opportunities to observe such dispersal directly are rare, so ecologists often turn to experimental methods to better understand the role of dispersal in limiting the distribution of species.

▼ Figure 52.18 Flowchart of factors limiting geographic distribution. An ecologist studying factors limiting a species' distribution might consider questions like these. As suggested by the arrows leading from the "Yes" responses, the ecologist would answer all of these questions because more than one factor can limit a species' distribution.



? How might the importance of various abiotic factors differ for aquatic and terrestrial ecosystems?